

# End to End FDE Project

From Customer Problem Statement to Production

Real Problem

Development

Production

# FORWARD DEPLOYED ENGINEER

The engineer who turns a customer problem into a working production solution

Customer-facing

Hands-on

Production

1

**Understand**  
Customer problem

2

**Build**  
Technical solution

3

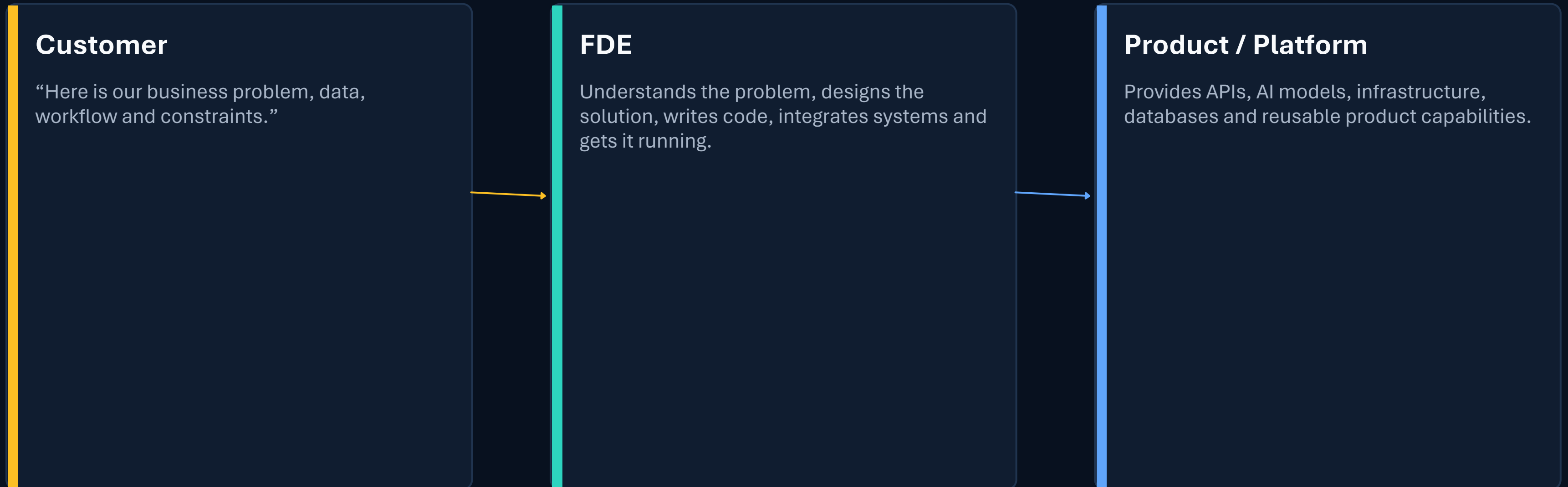
**Deploy**  
Into production

4

**Improve**  
With feedback

# What is a Forward Deployed Engineer?

Think of an FDE as a bridge between the customer and the technology team.



**The FDE translates business needs ↔ technical implementation.**

# The FDE Mission

Own the journey from “we have a problem” to “it works for real users.”



**Key idea: FDEs are measured by customer outcomes—not just by how much code they write.**

# Core Responsibilities

What does an FDE actually do day to day?

## 1. Understand requirements

Meet users, ask questions, map workflows, identify pain points.

## 2. Prototype quickly

Build a small working version to validate the idea.

## 3. Engineer the solution

Write production code, APIs, data pipelines, AI/RAG/agents, integrations.

## 4. Deploy & integrate

Connect customer systems, cloud, auth, databases and monitoring.

## 5. Troubleshoot

Debug failures, latency, bad outputs, data issues and edge cases.

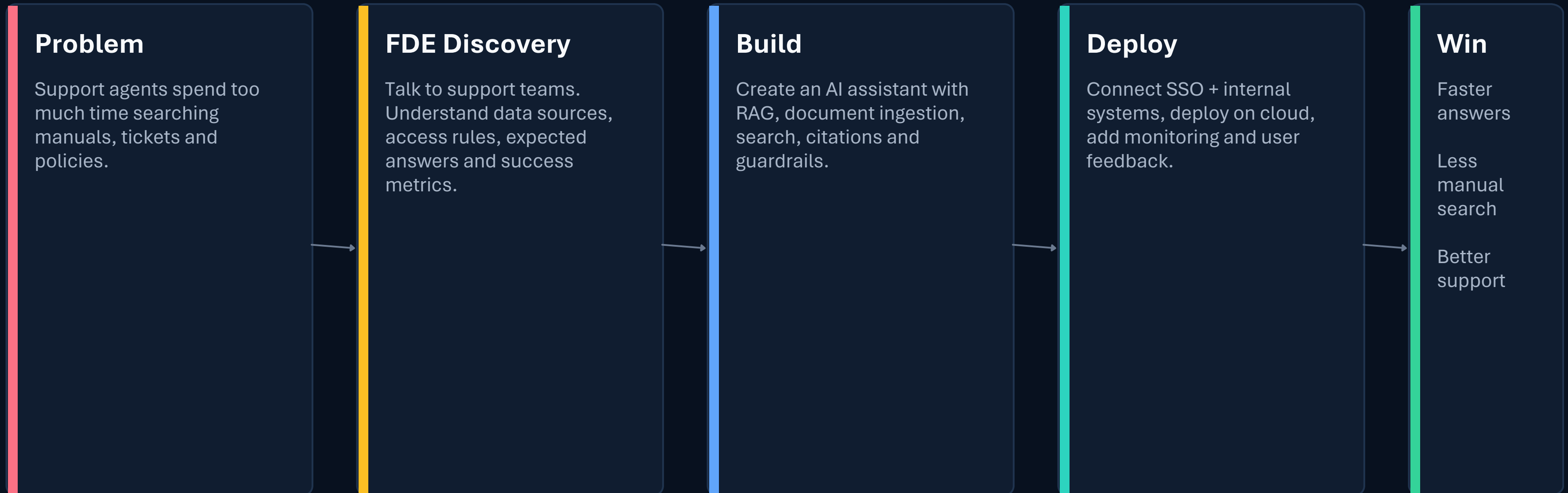
## 6. Iterate with users

Collect feedback, measure impact and improve the solution.



# A Simple Real-World Example

A support company has 100,000 internal documents. Employees cannot find answers quickly.



**FDE success = the customer can actually use it and gets measurable value.**

# What Makes an FDE Different?

The role combines multiple skill sets in one person.

- |   |                           |                                                           |
|---|---------------------------|-----------------------------------------------------------|
| 1 | <b>Software Engineer</b>  | Strong coding, APIs, systems, debugging                   |
| 2 | <b>AI / Data Engineer</b> | LLMs, RAG, agents, data pipelines, evaluation             |
| 3 | <b>Solution Architect</b> | Designs how systems connect and scale                     |
| 4 | <b>Consultant</b>         | Understands business context and asks the right questions |
| 5 | <b>Product Thinker</b>    | Prioritizes user value and fast iteration                 |

**FDE = Engineer + Architect + Problem Solver + Customer Partner**

# FDE vs Traditional Software Engineer

Both are engineers—but their operating environment is different.

## Traditional Software Engineer

- Often builds a core product
- Requirements usually come through product teams
- Works mostly inside the engineering organization
- Optimizes reusable product features
- Customer exposure varies

## Forward Deployed Engineer

- Builds for a specific customer problem
- Often discovers requirements directly
- Works closely with customer teams
- Optimizes time-to-value + production fit
- High customer interaction



# FDE vs Solutions Engineer / Consultant

These roles overlap, but the FDE is typically more implementation-heavy.

## Solutions Engineer

Demo + technical pre-sales

Proves the product can solve the problem

Often before / around the sale

## Consultant

Advice + transformation

Recommends processes and solutions

May or may not build the system

## FDE

Build + integrate + deploy

Owens a working customer outcome

Deep hands-on engineering

*Exact titles vary by company. Always read the job description.*

# Typical AI-FDE Technical Stack

You do not need every tool—but you need strong fundamentals across the stack.

## Application

Python • FastAPI • backend services

## AI

LLMs • prompting • RAG • agents • tool calling • evaluation

## Data

SQL • vector databases • ETL • document pipelines

## Integration

REST APIs • webhooks • OAuth/SSO • third-party systems

## Cloud / DevOps

Docker • CI/CD • cloud deployment • observability

## Engineering

Git • testing • security • debugging • system design

# The Most Important Non-Technical Skills

Technical ability gets you in. These skills make you effective with customers.

## Communication

Explain complex systems simply.

## Problem framing

Find the real problem behind the request.

## Ownership

Take responsibility from discovery to production.

## Speed

Prototype quickly without losing engineering judgment.

## Adaptability

Every customer environment is different.

## Trust

Be reliable with customer data, deadlines and expectations.

# FORWARD DEPLOYED ENGINEER

**Understand → Build → Integrate → Deploy → Improve**

The best FDE is not the person who knows every technology.  
It is the person who can understand a real problem, choose the right tools, and deliver a solution that works in production.

**Customer problem → Production outcome**